Email Phishing Analysis Report

Project: Offline Email Phishing Detection System
Analysis date: 15 October 2026
Test input: Synthetic sample_suspicious.eml

1. Objective

Identify phishing indicators from raw email headers and message content, calculate a heuristic risk score, and produce an auditable list of findings.

2. Email summary

Field	Observed value
Subject	URGENT! Verify your account immediately
From	PayPal Security <security@paypa1.com>
Return-Path	<alerts@unknown-site.example>
Received headers	1
DKIM signature present	False
Risk classification	HIGH - 100/100

3. Detection findings

+25 points: SPF reported failure

+25 points: DKIM reported failure

+30 points: DMARC reported failure

+10 points: From / Return-Path domain mismatch (may be legitimate)

+20 points: Urgent or coercive wording (7 patterns)

+10 points: Unencrypted HTTP link: paypal-login.example

+20 points: Possible brand impersonation in link: paypal-login.example

4. Authentication observations

The sample Authentication-Results reports SPF=fail, DKIM=fail, and DMARC=fail. These are values read from the supplied email, not independently validated by this tool. The From and Return-Path domains also differ; this can be legitimate in some email workflows.

5. Scoring method

The analyzer adds weighted points for reported authentication failures, sender-domain mismatch, missing header evidence, coercive phrases, and risky URL patterns. Scores are capped at 100. Classification: LOW (0-29), MEDIUM (30-59), HIGH (60-100). This is an explainable heuristic, not a calibrated probability.

6. Conclusion and recommended action

The synthetic test email is classified HIGH risk (100/100). Do not open the included link or provide credentials. Verify any claimed account alert through the organization's independently accessed official website or app.

7. Limitations

This tool does not independently query DNS, verify DKIM cryptography, validate SPF routing, inspect attachments, expand short links, or determine whether a destination is malicious. Received headers and Authentication-Results may be forged. Brand detection uses a limited example list and can generate false positives. A low score does not establish that an email is safe.

8. Reproduction

Run: `python email_analyzer.py sample_suspicious.eml --output sample_audit.json`

All domains and messages in the example are synthetic and intended for educational testing.